ENGINEERING | McMaster University

ENG1P13 Design Communication Workshop #6

Students' Handout

This workshop is to support your P3 Reflection Assignment. The activities in this handout are intended to be completed by the end of the Workshop #6. However, students are encouraged to apply what they learn in this workshop to their studies. The aim is to build a lifelong habit of critical reflection.

Objectives

By the end of this workshop, students should be able to:

- Define the concept of critical reflection
- Explain the importance of critical reflection
- Describe the 3-stage model of reflection
- Write a critical reflection on design thinking focusing on Project 3 experience

Summary of Activities and Lab Schedule

Workshop Introduction (5 min)

Introduction to Reflection, reflection model, "What"? (15 min)

Introduction to activity 1 (5 min)

Activity 1 (20 min)

So what? And intro to activity 2 (10 min)

Activity 2 (20 min)

Now what? And introduction to activity 3 (10 min)

Activity 3 (20 min)

One-on-one meeting with IAI/TAs (30 min)

Final Q/A and submissions (10 min): Following the lead of your TA or IAI, complete this handout during the workshop. At the end of the workshop, show your reflection to a TA or IAI for grading, and submit the handout as a single pdf file on Avenue to Learn. You are not required to write a formal reflection essay. This is an individual submission.

Upload your reflection essay as a **PDF** to the *Avenue Dropbox* titled **P3 Reflection** using the **MacID_P3_Reflection.pdf** as naming convention.

Your assignment is due at 23:59 EST the day of your scheduled lab. For example, if the lab is Monday at 08:30 EST, the assignment is due at 23:59 EST on Monday.

Grading of Reflection

Your reflection assignment is worth 0.5 mark of your total Project-3 grade (6.25%). Rubric is provided on Avenue to Learn.

If you need to review the content, go to the online reflection module.

Introduction

Through a written reflection, you will be guided to explore an experience of working on a team during your Project 3, focusing on the design process. To accomplish this task, you will be asked to identify, question, and assess your understanding of what happened. In this way, you make meaning of your experiences through this process of reconciling what you expected would happen with what occurred. Reflection is not a reading assignment, a summary of an activity, or an emotional outlet. Rather, reflection is a way for you to achieve a deeper sense of understanding of what you have learned.

In this workshop, you learn how to write a thoughtful and professional reflection, and you will practice it by reflecting on your Project 3 experience.

Reflection Activities

Consider your experience with the design process as a first-year engineering student working on Project 3 over the past couple of months in ENG 1P13. After exploring the client's challenges and gaining insights, your team decided to focus on one area to improve our client's daily life. You have defined the problem in a problem statement that included objectives, constraints, etc. Through this exploration, you performed a functional analysis that was used to come up with different alternative ways to solve the problem. Your team needed to make a decision between the different alternatives, and you tested your ideas for feasibility. You have been encouraged to iterate as you gained deeper insight and developed empathy for the client. Through the process of iteration, you have had the opportunity to improve upon your ideas. Engineers are continually iterating through the design process. Informed designers are involved in continual learning: learning by doing, learning from brainstorming and prototyping, learning by iteration and from feedback and failure, learning by noticing and troubleshooting, learning by drawing and dialoging about ideas, materials, and people. While iteration is an informal form of reflection, you will deepen your understanding of what you have learned through formal reflection. All of these emphasize the metacognitive and reflective practice aspects of learning through design (Lawson & Dorst, 2009; Crismond & Adams, 2012).

Activity #1: What?

In this section, you will describe a critical incident that you will be reflecting on as related to the "Generating/Testing Ideas" and "Decision-Making" stages of your design process.

In **five to seven sentences**, identify and describe **ONE critical incident**, breakthrough or big thought-provoking moment that either challenged your initial assumptions, had a positive impact on you or validated your understanding of the design process.

You can use the following lists of questions as inspiration to help you think of an experience on which you would like to reflect:

Generating & Testing Ideas:

- How did you go about exploring ideas?
- How deeply did you explore your design options?
  - How much research did you conduct?
  - Did you look into bio-imitation tools? How did these tools influence your idea generation?
  - Did you consider any "What if?" questions in your explorations?
- Did you test your ideas?
- If yes, how did you test your ideas?
  - What were you trying to test (e.g., desirability, feasibility, etc.)?
  - What tool/ method did you use? (physical prototype, CAD model, etc.)
  - How much time did you spend on testing each idea?
  - How many ideas did you test?
  - How many prototypes did you make for testing each idea?
  - Did you test your ideas early on or wait until you had more details of the ideas?
- What was one challenge that you faced in the testing process of the design? (we encourage you to write more than one challenge). And what did you do to solve that challenge? (you can attach photos to explain your attempted solutions)
- Ex: From the results of our testing, one change we made to improve our design solution was … (add your response) and this change made our design solution better because … (add your response).

Decision Making:

- What happened during decision-making?
  - Where in the process, relative to the design process steps, did you make decisions?
  - What were the decisions about? Decisions could be about the process (e.g., how much researching of the design space was enough?) or about the design (e.g., which design to prototype, deciding between alternative prototypes, etc.).
  - How many options did you have to choose from?

  - How many criteria did you have to compare the options? How did you choose those criteria?
  - What tools did you use to make a decision?
- At what stage did you make a decision?
- When did this experience take place? Did you already have one final solution in mind or you were still exploring the ideas?
- What challenges did you face during the decision-making process? At what point in the design process did these challenges happen?

To **describe the context** of your experience, consider the following questions:

- What happened?
- What did you do?
- What did you expect?
- What was different from what you expected?
- What went well?
- What went poorly?
- What was your reaction?

Your response:

As our group wanted to help Kimberly gain better spatial awareness, our initial idea involved creating neon-coloured NFC stickers that could be easily seen by her and could be scanned by her phone to verbally tell her what object she was detecting. This would allow her to better determine where different objects are in her surroundings via the neon colour of the stickers, accurately detect different objects she looks at with the NFC tags embedded into the stickers and would provide her with a more efficient strategy to organize her things. Unfortunately, during Kimberly's second Q&A session, our group learned that Kimberly was already very proficient in organizing and keeping track of her possessions and only had trouble recognizing new objects. Moreover, she also said that she could only see white-on-black or neon-on-black colour contrasts, which meant that our stickers would be ineffective in helping her notice where different objects are around her. This left our group in a predicament as we were already almost two milestones in before realizing that our design would be ineffective to her. Moreover, we already had two TA check-ins before this and had multiple points during the project where we could have stopped the construction of this design.

Stage 2: "So What?"

In this section you will explore what you learned and describe why this incident matters to you.

In **three to five sentences**, discuss what you learned from this incident about idea generation, testing ideas, and decision making that either surprised you, made you confront a misconception, or improved your understanding of the design process.

To help you think about this, consider the following:

- What was the outcome of early or late testing processes?
- Do you think delaying any of your decision-making may have improved the design?
- Could you have collected better observations or data that would have led to better decisions?
- Did you repeat your decision-making process at any other stage?

Your response:

From this incident, I learned that I need to test and question my design more in the future. The reason why our initial design has progressed so far was because we had not conducted enough testing on the logic of the design, and we also did not question various aspects of the design itself. Logical testing of the design would consist of walking through the steps of using the design and determining if its use would make the task the user is trying to accomplish more or less convenient. Moreover, our initial design was also insufficient because it tried to create a new way to solve the problem rather than trying to enhance or improve Kimberly's current method of recognizing objects, which could have made our design more effective and actively solve a problem Kimberly experienced.

In **two to three sentences,** explain why these new insights are important to you.

Your response:

These new insights into design testing and questioning are important to me because it provides new ways to make the design the problem more effectively and a new method of filtering out illogical and inefficient design ideas. By looking at the current method a problem is being solved or handled, I can create a more effective design by aiming to enhance the current solution rather than trying to reinvent it. Then, to test my design, I can walk through the steps of how the design logically works to see if the design has good logic and also solves the problem in an efficient way.

Stage 3: "Now What?"

In **two to three sentences**, discuss how you will integrate this new insight into future design projects. To help you think about this, consider the following:

- I learned that... (Express an important lesson, not a statement of fact)
- This learning matters because... (Consider how this learning has value to you as an engineer)
- How will I apply my learning?
- How will I design differently next time?
- How will I deal with a similar situation in the future?
- Considering this learning, I will... (Set specific, assessable goals; consider benefits and challenges involved in this plan)

Your response:

Within this design project, I learned that creating a design that enhances the current solution to a problem and testing the design's logical process are great ways to make a design more effective. In the future, I plan to conduct extensive research into current ways a problem is being solved and determine the core pitfalls as to why the current solution to the problem is ineffective. I will also create more flowcharts for my future designs to make it easier to test their logical flows and to determine their efficiency in solving the problem.

In **two to three sentences**, describe the possible benefits and challenges involved in your plan.

Your response:

While researching existing solutions to a problem, it is possible that the problem may not have any existing solutions. In this case, I would break up the problem into smaller problems and investigate how those problems are solved to inspire the solution to the overall problem. Moreover, not all design ideas may have a logical flow to them, meaning that they cannot be tested and must be tested in some other way before moving onto the next step in the design process.

References:

Lawson, B., & Dorst, K. (2009). Design expertise. Oxford, UK: Architectural Press.

Crismond, D. P., & Adams, R. S. (2012). The informed design teaching and learning matrix. Journal of Engineering Education, 101 (4), 738-797.

PROJECT THREE **REFLECTION – INDIVIDUAL** RUBRIC

Criteria	1 Below Expectation	2 Marginal	3 Meets Expectation	4 Exceeds Expectation
Communication	Many punctuation, grammar, and capitalization errors.	Noticeable punctuation, grammar, and capitalization errors.	Professionally written with minimal errors.	Professionally written with no errors. Good flow of ideas and concise when necessary.
Content – What?	Missing 3 or more items from Level 3.	Missing 1 or 2 items from Level 3.	• Written in first person • Clear description of context. i.e., history, environment, and/or key participants - who, what, when, where, how, why? • Draws from most senses to describe the incident, experience, problem, situation • Offers concrete examples and illustrations to clarify and enhance knowledge claims and understanding. • Uses some judgment, interpretation, and/or personal bias in explanations. • Identifies some initial thoughts or feelings experienced during the described event • Provides definitions of technical terms and key concepts.	
Content – So What?	Missing 3 or more items from Level 3.	Missing 1 or 2 items from Level 3.	• Identifies successes or positive aspects of the experience. • Identifies challenges that were experienced. • Identifies why the experience they chose to highlight was important to their learning. • Identifies how their knowledge or perspective has changed as a result of the experience.	
Content – Now What?	Missing all items from Level 3.	Missing 1 or 2 items from Level 3.	• Explicitly mentions something that they learned during the experience.	

			• Explains why their new learnings are important. • Outlines how they might use their new learnings in the future.	

List of Penalties	Deduction
Missing Name or MacID	**–10%**
Late Penalty	**–20% per day**